



EcoQuest Web Application | IIT Challenge 2026

Desa Marga Mulya

The Digital Gateway for a Village

Kenali desanya. Akses layanannya. Dukung ekonominya.

Connect residents, local businesses, and people outside the village through one digital gateway.



Member 1



Member 2

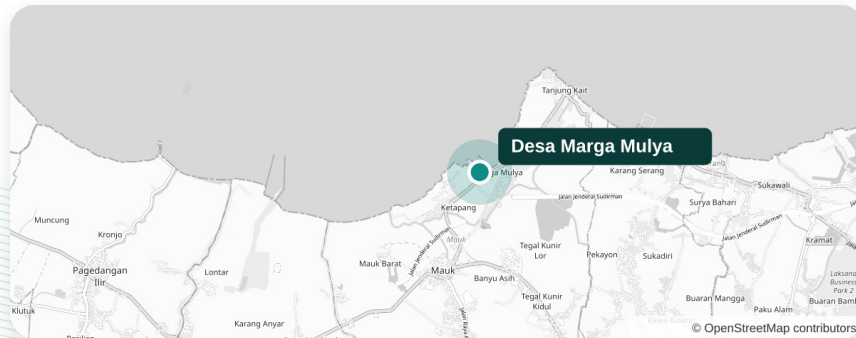


Member 3



A Coastal Farming Village

Marga Mulya, Mauk, Tangerang



7,842

residents

2,318

households

412 ha

village land

55%

of land is rice fields & fish ponds

41%

of workers are in farming & fisheries

Marga Mulya is a coastal village shaped by agriculture, fisheries, community, and a growing local economy.



Agriculture



Fisheries



Community



Local economy

Scattered, Not Missing

The village has information and potential — but it is hard to find, access, and use.



RESIDENT

1

“Who do I even ask about this?”

- Services & requirements
- Announcements & events
- Village information



BUYER OUTSIDE THE VILLAGE

2

“What can I buy from this village?”

- Local products
- Who sells them
- How to order



PARTNER / RESEARCHER

3

“Where is the data, and can I trust it?”

- Demographics
- Farming & fishery potential
- Facilities & location

The problem is fragmentation, not absence.

It lives in offices, chat groups, social media, and word of mouth — not in one place.

74%

rural residents online

6 / 10

MSMEs still offline

None

official village website found*

One Gateway, Three Audiences

Not just a village website — a digital gateway.



RESIDENTS

"I need something from the village."

- Services & checklists
- Live office status
- Announcements
- AI assistant



LOCAL BUSINESSES

"I want more customers."

- Pasar Desa marketplace
- Seller portal
- WhatsApp orders



PEOPLE OUTSIDE

"I want to know and support the village."

- Interactive map
- Explore by category (next)
- Open village data
- Local products

FOUR PILLARS



SERVE

- Live



DISCOVER

- Map live
- Categories next



BUY

- Live



UNDERSTAND

- Live

Built and Live

Live demo: desa-marga-mulya.vercel.app

Home

Service

Pasar Desa

Product

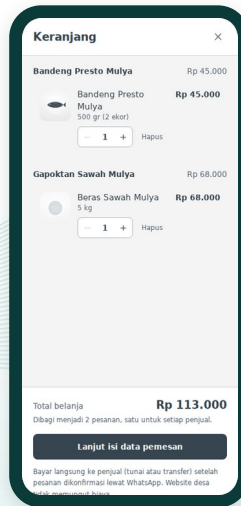
Cart

WhatsApp Order

Seller Portal



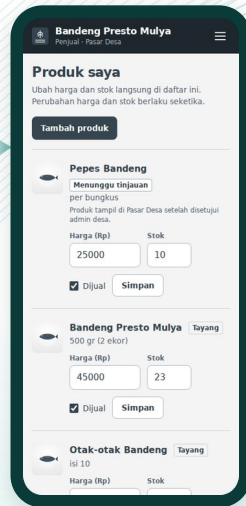
Home — live office status & services



Multi-seller cart



WhatsApp order



Seller portal

Also live: service checklists · Tanya Desa AI assistant · 42 open data tables with CSV · interactive map · CMS for village staff

A Practical Win-Win-Win

Impact & implementation feasibility



RESIDENTS

24/7

information
access

- ✓ Easier access to services
- ✓ Fewer unnecessary trips
- ✓ 24/7 information access



BUSINESSES

0%

platform commission

- ✓ 0% platform commission
- ✓ Digital visibility
- ✓ Simple WhatsApp ordering



VILLAGE

1

hub for village information

- ✓ Centralized information
- ✓ CMS-managed content
- ✓ Open village data

**Feasible
today**



Rp 0

monthly hosting on the
current free-tier setup



PostgreSQL

reliable, managed database



CMS

11 content collections, no
coding needed



109

automated checks, all passing

One Gateway for the Whole Village



Residents access services.



Businesses reach customers.



People outside discover and support Marga Mulya.

Kenali desanya. Akses layanannya. Dukung ekonominya.

10

public pages

42

data tables

16

local products



Scan to try the live demo

desa-marga-mulya.vercel.app

Thank you